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FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

ECONOMICS 114 A1

Date:	8 April 2025
Duration:	2 hours
Total Marks:	60
Examiners:	Dr C. Links (Convenor) Dr K. Malindi Dr M. Nchake Mr J. Pretorius Ms L. Dube Ms R. Petersen Ms J. Kazumunu
Internal Moderator	Dr M van Schoor

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.
ANSWER SECTION A's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY MAKING A CROSS NEXT TO THE NUMBER OF THE STATEMENT.
ALSO ANSWER SECTION B's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY MAKING A CROSS NEXT TO THE NUMBER OF THE STATEMENT.
ANSWER SECTION C's QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS.

Key Abbreviation	Meaning	Key Abbreviation	Meaning
AP	Average product	BL	Budget line
GDP	Gross domestic product	IC	Indifference curve
K	Capital	L	Labour
FF	Feasible frontier	MP	Marginal product
MRS	Marginal rate of substitution	MRT	Marginal rate of transformation
P	Price	PF	Production function
Q	Quantity	TP	Total product
w	Wage	Y	Income

SECTION C

DESCRIPTIVE QUESTIONS

Question 19 [3 marks]

The recent announcement of the Expropriation Bill, which legalises certain types of land expropriation (without compensation) under specific conditions in the public interest, has sparked widespread controversy in South Africa. If such a policy were to be implemented recklessly, what impact might it have on:

19.1 Economic institutions in South Africa?

Comment: Many students found it challenging to articulate the relationship between capitalist institutions and private property rights, particularly in the context of the potential consequences of poorly implemented expropriation legislation.

19.2 Investment in South Africa?

Comment: This question was generally well answered, with several students correctly linking poorly implemented expropriation legislation to a decline in both domestic and foreign investment. However, a notable number of students mistakenly assumed that such a policy would lead to increased investment levels, reflecting a misunderstanding of investor behaviour under institutional uncertainty.

19.3 Long-term economic growth?

Comment: Most students were able to identify the adverse impact that the reckless implementation of this policy could have on long-term economic growth. Many correctly recognised that heightened uncertainty would discourage firms from investing, thereby undermining growth prospects.

Question 20 [7 marks]

The following table shows the input costs of two different technologies (Technology A and Technology B) used to produce 1 000 bottles of sparkling wine. Assume that the production of sparkling wine requires only two inputs: grapes and labour.

Technology	Grapes (kg)	Labour (hours)
A	y	800
B	800	1 200

20.1 Labour costs R8 per hour, grapes cost R10 per kilogram, and the total cost of Technology A is R18 400.

20.1.1 If grapes are represented on the vertical (y-) axis, derive the equation of the isocost line corresponding to each of the two technologies.

[2]

Comment:
This question was particularly poorly answered. Approximately half of the students who attempted A1 were unable to derive the correct equations for the isocost lines, namely $1840 - 0.8x = y$ and $1760 - 0.8x = y$. The presence of an unknown in the table appears to have significantly hindered students' ability to formulate the equations correctly.

20.1.2 Which technology will the sparkling wine producer choose? Why?

[1]

Comment: This question was answered well. Almost all students could correctly identify that technology B was the cheapest at R17 600

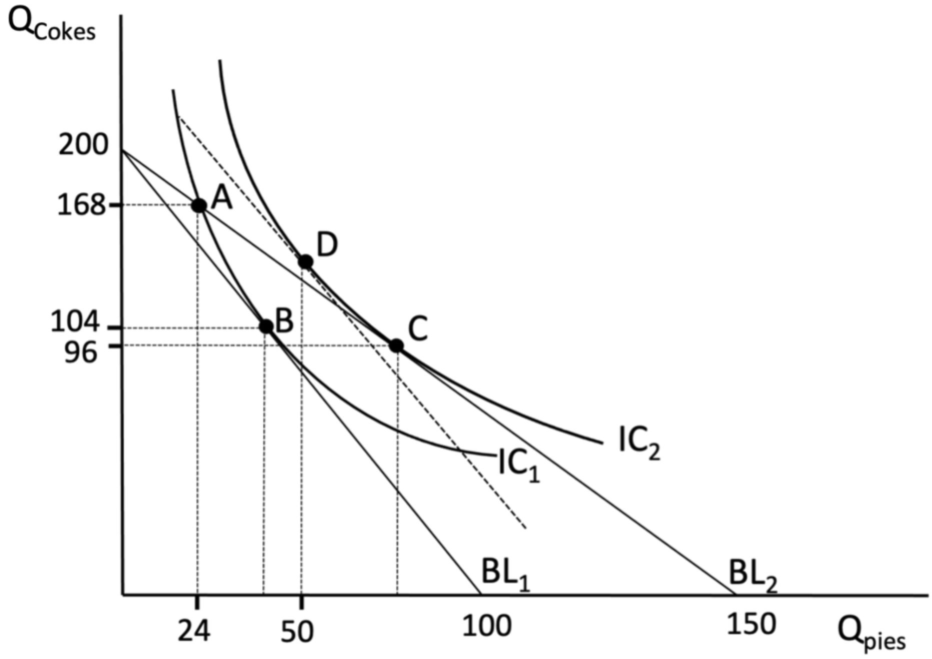
20.2 Suppose that only the relative price of a kilogramme of grapes changes to the equivalent of 2 labour hours — which technology will the winemaker choose now?

[4]

Comment: This question was generally poorly answered, with only a small proportion of students arriving at the correct solution. A common error involved the incorrect assumption that the doubling of the price of labour to R16 implied a new, higher price for a kilogram of grapes. Many students erroneously concluded that R16 was the new price of grapes, overlooking the change in relative prices. In fact, the correct monetary price of a kilogram of grapes falls to R5.
On a more positive note, most students were able to correctly compute the value of y = 1200, indicating a sound understanding of the relevant algebraic manipulation. However, many failed to link this to the cost comparison between the two technologies. The correct approach reveals that the total cost of Technology A becomes R12,400, while Technology B rises to R13,600. Therefore, in the revised cost scenario, Technology A becomes the more cost-effective option.

Question 21 [5 marks]

The figure below (not drawn to scale) shows two budget lines, BL_1 and BL_2, as well as two indifference curves for Calvino, a consumer of Coke and pies. Calvino spends a total of R3 000 on these two goods. A decrease in the price of pies shifts Calvino's optimal consumption from point B to point C.



21.1 The impact of a decrease in the price of pies can be broken down into two components: (i) the income effect and (ii) the substitution effect. Use the graph above to explain how these two effects contribute to the overall change in consumption.

[3]

Comment:

- This question was generally well answered, as most students correctly identified the movement from point B to point D as representing the income effect. This indicates a good conceptual grasp of the basic mechanics of consumer choice. However, a significant number of students did not acknowledge that, in the case of the income effect, relative prices are held constant. Specifically, many failed to explain that the decline in the price of pies is interpreted as an increase in real income, which alters the consumer's purchasing power without changing relative price ratios. This omission suggests a partial understanding of the distinction between income and substitution effects.
- Most students were also able to correctly identify the substitution effect, represented by the movement from point D to point C. They appropriately recognised that this effect arises because pies become relatively cheaper following a change in price. In

line with standard consumer theory, students generally understood that this relative price change induces the consumer to substitute away from the now relatively more expensive good (Coke) toward the relatively cheaper good (Pies). Furthermore, many responses accurately noted that this substitution occurs while holding utility constant at the new indifference curve, thereby isolating the substitution effect from the income effect. This indicates a solid conceptual understanding of how changes in relative prices influence consumption choices in microeconomic models.

21.2 Does the income or substitution effect dominate for Coke? Explain your answer.

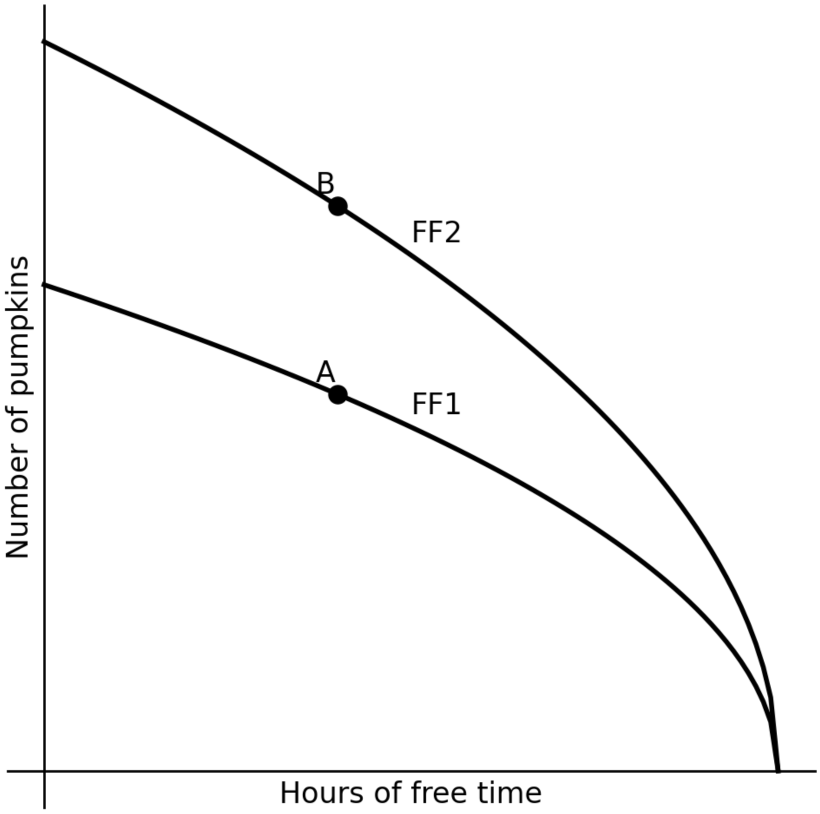
[2]

Comment:

This question was generally poorly answered. Only a small number of students were able to clearly articulate that the diagram illustrates a case in which the substitution effect for Coke outweighs the income effect. Many responses did not interpret the graphical representation correctly or apply the theoretical framework that distinguishes between income and substitution effects. This suggests a gap in students' ability to evaluate the relative magnitudes of these effects using visual or theoretical cues.

Question 22 [5 marks]

The graph below illustrates two feasible frontiers for Palesa, a pumpkin farmer.



22.1 Explain the economic forces that could cause the shift from FF_1 to FF_2?

[3]

Comment:
Approximately 60% of students were able to answer this question correctly. Many encountered difficulty in explaining that the shift from point A to point B on the feasible frontier was due to productivity gains, and specifically in defining those gains in terms of increased pumpkin output. While students were generally more successful in listing plausible reasons for such a shift—such as technological improvements or enhanced labour efficiency—they often failed to connect these reasons to the underlying concept of productivity in a quantifiable manner. This pattern indicates a common issue: while students are able to recall examples or explanations at a surface level, they struggle with applying core analytical concepts such as productivity to interpret graphical shifts. The responses suggest that there is a gap between rote learning and a deeper conceptual understanding of the feasible frontier and its dynamics as presented in Unit 3.

22.2 How does this shift affect the marginal rate of transformation (MRT) of free time into pumpkins and why?

[2]

Students generally performed well on this question, with many correctly identifying that the marginal rate of transformation (MRT) becomes steeper as a result of productivity gains. They were able to explain that these gains alter the trade-off between pumpkins and free time—specifically, that more pumpkins must now be given up to gain an additional hour of free time, hence a steeper MRT. However, students who struggled with this question often demonstrated confusion between the concepts of the marginal rate of substitution (MRS) and the marginal rate of transformation (MRT).

Question 23 [5 marks]

Consider the game represented in the table below:

		Country B	
		Free Trade	**Impose Tariffs**
Country A	**Free Trade**	2, 2	0, 3
	Impose Tariffs	3, 0	-5, -5

23.1 Find the Equilibrium solution(s) for the game and identify what type of game is represented in the matrix.

[3]

This question was generally poorly answered, which is unfortunate given that it offered students a valuable opportunity to earn marks on the Unit 4 game theory material. While a number of students were able to correctly identify that the game had two Nash equilibria, they often failed to substantiate their answers by explicitly stating the associated payoffs—(3,0) and (0,3). Furthermore, a common error involved misclassifying the structure of the game. Instead of identifying it correctly, many students labelled it as an ultimatum game, a prisoner's dilemma, or even an invisible hand game. These misidentifications suggest a lack of conceptual clarity in distinguishing between the different types of strategic interactions introduced in the course. The overall performance on this question points to the need for additional reinforcement of key concepts in game theory, including payoff structures and the defining features of different canonical games.

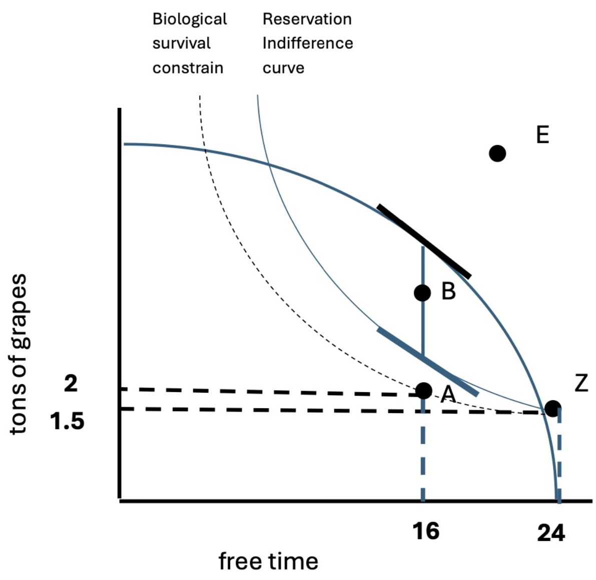
23.2 Do any players have a dominant strategy? Please justify your answer?

[2]

This was very well answered as students could easily demonstrate with there answered from 23.1 that neither Country A nor Country B had a dominant Strategy in this game

Question 24 [5 marks]

Consider the diagram below:



24.1 Which combination of grapes per ton and free time (represented by points A, B, E, and Z) is infeasible?

[1]

Almost all students correctly identified point E, demonstrating a sound understanding of the feasible set. However, a significant number of students also incorrectly identified point Z as a feasible choice. This is a conceptual error, as point Z represents a situation in which the individual does no work but still receives government rations. Such a scenario falls outside the boundaries of the feasible set defined by the model, which assumes that consumption must be earned through productive labour. The inclusion of point Z as a viable option reflects a misunderstanding of the constraints imposed by the feasible frontier and the underlying assumptions of the model. This indicates the need to reinforce the distinction between hypothetical points presented for illustrative purposes and those that are genuinely feasible given the economic constraints in the model.

24.2 If Johan offers Lisa a wage of 1.5 tons of grapes, how many hours per day would she choose to work? Explain your reasoning.

[2]

Students were generally able to identify correctly that Lisa should choose to work zero hours per day. However, most students struggled to provide a coherent explanation for this decision. Specifically, they did not recognise that at the prevailing ration level, Lisa would be earning exactly her reservation wage—the minimum compensation required for her to be indifferent between working and not working. As a result, it would be rational for her to remain at home, since the opportunity cost of working outweighs any additional benefit. This oversight suggests that while students can often identify the correct outcome in labour-leisure choice scenarios, they may lack a deeper conceptual understanding of the role of the reservation wage in shaping labour supply decisions.

24.3 At point A, if Johan offers Lisa a wage of 2 tons of grapes and expects her to work 8 hours per day, should she accept this offer? Explain your reasoning.

[2]

Many students were able to identify correctly that Lisa would reject the take-it-or-leave-it offer of 2 tons of grapes in exchange for 8 hours of work. However, only a small number of students could adequately explain the reasoning behind her decision. In particular, few recognised that the offer would leave Lisa with just enough to meet her subsistence needs, without providing any surplus. Consequently, she would not be indifferent between working and not working, as the utility gained from accepting the offer does not exceed the utility she would receive from remaining unemployed and surviving through other means. This reflects a broader difficulty among students in applying the concept of indifference and subsistence constraints to labour market coercion scenarios.g